

Name: _____

Draw a right triangle for each problem. Round as directed.

1. A ranger 85 ft up a tower spots smoke at an angle of depression of 6° . Find the horizontal distance to the smoke (nearest foot).
2. A flagpole casts a 74-ft shadow when the angle of elevation of the sun is 63° . Find the height of the flagpole (nearest foot).
3. A jet flies at 5200 ft. The angle of elevation from the runway to the jet is 24° . Find the horizontal distance (nearest foot).
4. From the rim of a canyon, the angle of depression to the base of the opposite wall, 130 ft away horizontally, is 43° . How deep is the canyon (nearest foot)?
5. A lifeguard 66 ft up a tower sees two swimmers in line with the tower, at angles of depression 54° (near) and 19° (far). How far apart are they (nearest foot)?
6. A zip line is anchored to the ground 42 ft from the base of a platform, and the cable makes a 37° angle with the ground. Find the platform height (nearest foot).
7. A loading ramp rises 28 in. over a horizontal run of 45 in. Find its angle of elevation (nearest tenth of a degree).
8. Explain why an angle of depression from A to B always equals the angle of elevation from B to A.
9. In problem 5, why is the FAR swimmer's angle of depression smaller?
10. A hiker sees a cliff top at 35° elevation. Walking closer, does the angle increase or decrease? Why?